Tuesday, October 6, 2026 2:35 PM

$$
\begin{cases}
y = X\beta + \varepsilon \\
\beta \sim N_p(\beta_0, V_0). \\
\varepsilon \sim N(0, \sigma^2 I_n). \\
\sigma^2 \sim IG(a_0, b_0)
\end{cases}
$$

$$
\begin{cases}
Y = X\beta + \varepsilon. \\
\varepsilon \sim N(0, \sigma^2 I_n) \\
p(\beta, \sigma^2) \propto \frac{1}{\sigma^2}
\end{cases}
\Leftarrow
\begin{array}{cc}
y_1 & X_1 \\
\vdots & \vdots \\
y_n & X_n
\end{array}
$$

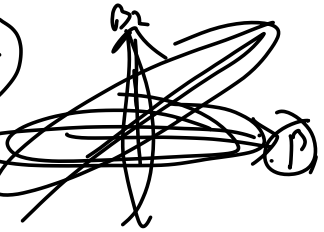
Likelihood: $L(\underline{\beta}, \underline{\sigma^2} | \underline{y}) = \frac{1}{(\sigma^2)^{\frac{n}{2}}} \exp\left\{-\frac{(y - X\beta)^T (y - X\beta)}{2\sigma^2}\right\}$

Posterior $\pi(\beta, \sigma^2 | \text{Data}) \propto$ $(\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{(y-X\beta)^T(y-X\beta)}{2\sigma^2}\right\}$

Check (with improper prior)

The posterior is proper iff $\int_0^\infty \int_{\mathbb{R}^p} (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{(y-X\beta)^T(y-X\beta)}{2\sigma^2}\right\} d\beta\, d\sigma^2 < \infty$.

$(X^TX + \lambda I)^{-1} X^T y$



$y = X\beta + \varepsilon$

$\hat{\beta} = (X'X)^{-1} X'Y.$

$\widehat{Var}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1}$

$\pi(\beta, \sigma^2) \propto \frac{1}{\sigma^2}$ $\pi(\beta_1) \propto 1$

$\pi(\beta,\sigma^2) = \pi(\sigma^2)\pi(\beta|\sigma^2)$ Prior $\beta_1 \sim N(0,\tau^2)$

$\pi(\sigma^2) = \frac{1}{\sigma^2}$ & $\pi(\beta|\sigma^2) = k$

$$Z = \int_0^\infty \int_{\mathbb{R}^p} (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{(y-X\beta)^T(y-X\beta)}{2\sigma^2}\right\} d\beta\, d\sigma^2 < \infty ?$$

When $\text{rank}(X) < p$, then is $z = \infty$? X^TX is not invertible.

" " $= p$ ——— $z < \infty$ if $\text{rank}(X) < p$

$$\int_0^\infty (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{y^Ty}{\sigma^2}\right\} \int \exp\left\{-\frac{(\beta^TX^TX\beta - 2\beta^TX^Ty)}{2\sigma^2}\right\} d\beta\, d\sigma^2.$$

If $\text{rank}(X) < p$.

Then, $\exists$ a non zero vector v such that $Xv = 0$



Let $u_1 = \frac{v}{\|v\|}$

Complete $(u_1, u_2 .. u_p)$ to be an orthonormal basis & $U = (u_1 \ldots u_p)$

Since U is orthogonal, $U^TU = I_p$.

$\forall \beta \in \mathbb{R}^p$ $\beta = Ut$ where $t \in \mathbb{R}^p$.

$$z = \int_0^\infty \int_{\mathbb{R}^p} (\sigma^2)^{(-\frac{n}{2}+1)} \exp\left\{-\frac{S(Ut)}{2\sigma^2}\right\} dt\, d\sigma^2$$

$X\beta = X(\beta + wv) \forall w$

$\Rightarrow \forall \|y - X\beta\| = \|y - X(\beta + wv)\|$

Define $S(\beta) = \|y - X\beta\|^2$.

$= (\infty (\int^\infty (\sigma^2)^{-(\frac{n}{2}+1)} \cdot (-S^*(t_2 - t_p) \ldots$

$$\int_0^\infty \int_{\mathbb{R}^{p-1}} \int_{-\infty}^{\infty} (\sigma^2)^{-\left(\frac{n}{2}+1\right)} \exp\left\{-\frac{S(t)}{2\sigma^2}\right\} dt_1 \, dt_2 \cdots dt_p \, d\sigma^2 = S^*(t_2, \ldots, t_p)$$

$$= \int_0^\infty \int_{\mathbb{R}^{p-1}} (\sigma^2)^{-\left(\frac{n}{2}+1\right)} \exp\left\{-\frac{S^*(t_2,\ldots,t_p)}{2\sigma^2}\right\} dt_1 = \infty.$$

$\therefore$ rank$(X) < p \Rightarrow$ No proper posterior

When rank$(X) = p$, then is $\pi < \infty$?

$$\int_{\mathbb{R}^p} \exp\left\{-\frac{S(\beta)}{2\sigma^2}\right\} d\beta. \qquad S(\beta) = (y - X\hat{\beta})^T (y - X\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta}).$$

where $\hat{\beta} = (X^TX)^{-1}X^Ty$.

$$= \exp\left\{-\frac{SSE}{2\sigma^2}\right\} |X^TX|^{-1/2} (2\pi\sigma^2)^{p/2}.$$

Let $SSE = (y - X\hat{\beta})^T (y - X\hat{\beta})$

$$\int_0^\infty (\sigma^2)^{-\left(\frac{n}{2}+1\right)} (\sigma^2)^{p/2} \exp\left\{-\frac{SSE}{2\sigma^2}\right\} d\sigma^2.$$

$$= \int_0^\infty \sigma^{2-\left(\frac{n-p}{2}+1\right)} \exp\left\{-\frac{SSE}{2\sigma^2}\right\} d\sigma^2.$$

This is the pdf of Inverse Gamma upto a constant.

Now IG pdf doesn't integrate to infinity

$$(IG(a_0, b_0)) \propto (\sigma^2)^{-(a_0+1)} \exp\left(-\frac{b_0}{\sigma^2}\right) \quad a_0 > 0, \; b_0 > 0.$$

$\Rightarrow$ $n > p$ & $SSE > 0$, rank$(X) = p$ are needed for proper posterior

$$\left.\begin{aligned} \beta | \sigma^2, y &\sim N_p(\hat{\beta}, \sigma^2 (X'X)^{-1}) \\ \sigma^2 | y &\sim IG\left(\frac{n-p}{2}, \frac{SSE}{2}\right) \end{aligned}\right\} \rightarrow \text{Matches classical regression results with this prior.}$$

Don't calculate $\beta | y \sim t_{n-p}\left[\hat{\beta}, \frac{SSE}{n-p}(X^TX)^{-1}\right]$

Prior as regularisers.

$y = X\beta + \varepsilon$.
$\varepsilon \sim N(0, \sigma^2 I_n)$.
σ^2 is fixed.
Proper prior.

$p(\beta|y) \propto p(y|\beta)\pi(\beta)$.

$\hat{\beta}_{MAP} = \arg\max_{\beta} p(\beta|y)$.
$= \arg\max_{\beta} \log(p(\beta|y))$
$= \arg\max_{\beta} \{\log p(y|\beta) + \log(\pi(\beta))\}$
$= \arg\min_{\beta} \{-\log p(y|\beta) - \log(\pi(\beta))\}$

For the linear regression likelihood,

$-\log(p(y|\beta)) = \frac{1}{2\sigma^2}\|y - X\beta\|^2 + C$

$\therefore \hat{\beta}_{MAP} = \arg\min_{\beta} \{\frac{1}{2\sigma^2}\|y - X\beta\|^2 - \log \pi(\beta)\}$

Penalty in regularised regression is the negative log-prior.

$\beta_j \sim N(0, \tau^2)$.

$\text{Min}\{\|y - X\beta\|^2 + \lambda\|\beta\|^2\}$.

$\lambda\beta_1^2 + \lambda_2\beta_2^2 + \lambda_3\beta_3^2$
$\lambda\beta_1^2 + \lambda\beta_2^2 + \lambda\beta_3^2$.

$\text{Min}\{\|y - X\beta\|^2 + \lambda\|\beta\|_1\}$.

standardisation is needed.

$\frac{x_1}{s.d(x_1)}$ $\frac{x_2}{s.d(x_2)}$

A1	Kg	Km.
A2	g	Km

First standardise, then.
have $\beta_1, \ldots \beta_p \sim N(0, \tau^2)$
Then Find the equivalence of ridge with β_{MAP}.
when $\beta_1 \ldots \beta_p \sim N(0, \tau^2)$

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MAP estimator for linear regression.

RIDGE

Match

LASSO

~~Fixed and~~ known σ^2.

Ridge: $\arg\min_{\beta} \left\{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right\}$ — (1)

$y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$. $\beta_j \overset{iid}{\sim} N(0, \tau^2)$ $j = 1 \dots b$

$\hat{\beta}_{MAP} = \arg\min_{\beta} \left\{ \frac{\|y - X\beta\|^2}{\sigma^2} + \frac{1}{\tau^2} \|\beta\|^2 \right\}$

$= \arg\min_{\beta} \left\{ \|y - X\beta\|^2 + \frac{\sigma^2}{\tau^2} \|\beta\|^2 \right\}$ — (11)

What prior (or what τ) will you choose to get a given value of λ for Ridge regression.

$\tau^2 = \frac{\sigma^2}{\lambda}$ or $\tau = \frac{\sigma}{\sqrt{\lambda}}$

LASSO (known σ): $\arg\min_{\beta} \left\{ \|y - X\beta\|^2 + \lambda_L \|\beta\| \right\}$ — (3)

What will be the Bayesian model (prior of β) such that

$\pi(\beta)$: prior of β $\hat{\beta}_{MAP} = \arg\min_{\beta} \left\{ \frac{\|y - X\beta\|^2}{\sigma^2} - \log \pi(\beta) \right\}$ (equivalent to (3))

$\sigma^2 \cdot \log \pi(\beta) = -\lambda_L \|\beta\|$.

$\Rightarrow \log \pi(\beta) = -\frac{\lambda_L}{\sigma^2} \|\beta\|$. $\Rightarrow \pi(\beta) \propto \exp\left(-\frac{\lambda_L}{\sigma^2} \|\beta\|_1\right)$

$\beta_j \overset{iid}{\sim} \text{Laplace}\left(\frac{\sigma^2}{\lambda_L}\right)$ $j = 1, 2, \dots b$.

$\|\beta\|_1 = \sum_{j=1}^{b} |\beta_j|$

σ^2 unknown: Ridge: $\arg\min \left\{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right\}$.

$y = X\beta + \varepsilon$

$\varepsilon \sim N(0, \sigma^2)$

$\beta \sim N(0, \tau^2)$

$\pi(\sigma^2) \propto \frac{1}{\sigma^2}$

$(\sigma^2, \beta)_{MAP} = \arg\max_{\sigma, \beta} \left\{ \pi(\sigma^2) \pi(\beta) \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ -\frac{\|y - X\beta\|^2}{2\sigma^2} \right\} \right\}$

$= \arg\min_{\sigma, \beta} \left\{ -\log(\pi(\sigma^2)) + n \log(\sigma\sqrt{2\pi}) - \frac{\|y - X\beta\|^2}{2\sigma^2} \right.$

$\|\beta\|^2$)

$\beta|\sigma \sim N(0, \frac{\sigma^2}{\tau^2})$

$\times \pi(\sigma^2)$

$= \arg\min_{\sigma,\beta} \left\{ C(\sigma) - \frac{\|y-X\beta\|^2}{2\sigma^2} - \frac{\|\beta\|^2}{2\tau^2} \right\}$

MAP for this model gives you the exact answer for β as Ridge Regression

$y = X\beta + \varepsilon$,

$\varepsilon \sim N(0, \sigma^2 I_n)$

$\sigma^2 \sim \pi(\sigma^2)$

$\beta|\sigma^2 \sim N(0, \frac{\sigma^2}{\tau^2} I)$

$\arg\min_{\sigma,\beta}$ (For every value of σ the loss fn. w.r.t β minimizes.)

$\min \{ \|y-X\beta\|^2 + \lambda \|\beta\| \}$ $\|y-X\beta\|^2 + \lambda\|\beta\|^2$ $\|y-X\beta\|^2 + \lambda\|\beta\|$

$C(\sigma) e^{-\frac{Q(\beta)}{\sigma}}$

$y = X\beta + \varepsilon$

$\varepsilon \sim N(0, \sigma^2 I_n)$

$\sigma^2 \sim \pi(\sigma^2)$

$\beta|\sigma^2 \sim \text{Laplace}(0, \frac{\sigma^2}{\lambda})$

Elastic net

$p(\beta|\sigma^2) \propto \exp\left\{ -\frac{(\lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2)}{2\sigma^2} \right\}$ $\hat{\beta} = \arg\min \{ \|y-X\beta\|^2 + \lambda\|\beta\|_1 + \lambda_2\|\beta\|^2 \}$

Bayesian Lasso

$y|\beta,\sigma^2 \sim N_n(X\beta, \sigma^2 I_n)$

$\beta_j|\sigma \sim \text{Laplace}(0, \frac{\sigma}{\lambda})$

$p(\beta_j|\sigma,\lambda) \sim \frac{\lambda}{2\sigma} \exp\left(-\frac{\lambda|\beta_j|}{\sigma}\right)$

Use this property to find Gibbs Sampler

If $\tau_j^2|\lambda \sim \text{Exp}(\frac{\lambda^2}{2})$

& $\beta_j|\tau_j^2,\sigma^2 \sim N(0, \sigma^2\tau_j^2)$

then, $\beta_j|\sigma,\lambda \sim \text{Laplace}(0, \frac{\sigma}{\lambda})$

(M) Normal & scaled-inverse-χ^2

$\sigma \sim (ik)$

unconditional it was.

$\mu \sim t$

$y|\beta,\sigma^2 \sim N(X\beta, \sigma^2)$

$\beta_j|\tau_j^2,\sigma^2 \overset{ind}{\sim} N(0, \sigma^2\tau_j^2)$

$\tau_j^2 \overset{iid}{\sim} \text{Exp}(\lambda^2/2)$

$\sigma^2 \sim \text{IG}(a_0, b_0)$

$\beta|\sigma^2, y, \tau^2 \sim$ [ ]

$\sigma^2|\beta, y, \tau^2 \sim$ [ ]

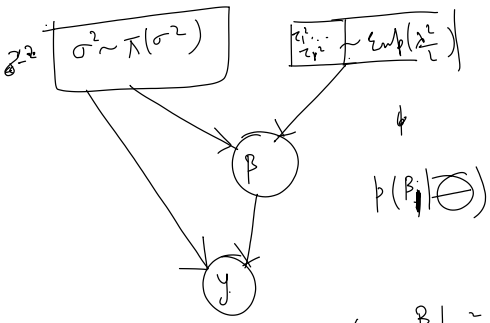
$\tau^2|\sigma^2, \beta, y \sim$ [ ]

Joint posterior: $p(\sigma^2, \tau^2, \beta|y) \propto p(\sigma^2, \tau^2, \beta, y)$

$= p(\sigma^2)\, p(\tau^2)\, p(\beta|\tau^2, \sigma^2)\, p(y|\beta, \sigma^2)$

$= p(\sigma^2) \left(\prod_{j=1}^{p} p(\tau_j^2)\right) \left(\prod_{j=1}^{p} p(\beta_j|\tau_j^2, \sigma^2)\right) \prod_{i=1}^{n} p(y_i|\beta, \sigma^2)$

$\rightarrow (*)$



$p(\beta_j|\ominus) \propto p(*)$

$\propto p(\beta_1|\tau_1^2, \sigma^2)\, p(y|\beta, \sigma^2)$

$\beta|\sigma^2 \sim N_p(V_\beta X^T y, \sigma^2 V_\beta)$ where $V_\beta = (X^TX + D_\tau^{-1})^{-1}$

where $D_\tau = \begin{pmatrix} \tau_1^2 & 0 & & \\ 0 & \tau_2^2 & 0 & 0 \\ & 0 & \ddots & \\ & & & \tau_p^2 \end{pmatrix}$

$\sigma^2|\beta, \tau^2, y \sim \text{IG}(a_n, b_n)$

$\tau_j^2|\sigma^2, \beta, y \sim$ [ ] (Metropolis-Hastings in this step)

MH within Gibbs